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# Airfoil Aeroelastic Flutter Analysis Based on Modified Leishman-Beddoes Model at Low Mach Number

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## Abstract

Based on modified Leishman-Beddoes (L-B) state space model at low Mach number (lower than 0.3), the airfoil aeroelastic system is presented in this paper. The main modifications for L-B model include a new dynamic stall criterion and revisions of normal force and pitching moment coefficient. The bifurcation diagrams, the limit cycle oscillation (LCO) phase plane plots and the time domain response figures are applied to investigating the stall flutter bifurcation behavior of airfoil aeroelastic systems with symmetry or asymmetry. It is shown that the symmetric periodical oscillation happens after subcritical bifurcation caused by dynamic stall, and the asymmetric periodical oscillation, which is caused by the interaction of dynamic stall and static divergence, only happens in the airfoil aeroelastic system with asymmetry. Validations of the modified L-B model and the airfoil aeroelastic system are presented with the experimental airload data of NACA0012 and OA207 and experimental stall flutter data of NACA0012 respectively. Results demonstrate that the airfoil aeroelastic system presented in this paper is effective and accurate, which can be applied to the investigation of airfoil stall flutter at low Mach number.

**Keywords:** low Mach number; unsteady aerodynamics; airfoil aeroelastic system; stall flutter; bifurcation

## 1. Introduction

Stall flutter is a nonlinear aeroelastic phenomenon, which commonly leads to failure of aeroelastic structure. The flow field during the dynamic stall process is largely separated. The physical process involves the formation of a vortex near the airfoil leading edge and shedding from the airfoil surface. The effect of dynamic stall enhances the complexity of the airfoil aeroelastic system. It makes the flutter behavior of airfoil aeroelastic system much more difficult to analyze. Stall flutter is a limiting factor of the performance of

fixed wing aircraft, helicopter and turbomachines etc., and research related to this issue is very plentiful<sup>[1-9]</sup>.

A prerequisite to stall flutter analysis is the accuracy of model for the prediction of unsteady and dynamic stall behavior. Recently, some semi-empirical unsteady and dynamic stall models are applied to the analysis of airfoil aeroelastic stall flutter, and Leishman-Beddoes (L-B) model<sup>[10]</sup> is chosen in this paper because of its wide engineering applications. Based on the L-B model, Beddoes<sup>[11]</sup> and Galbraith<sup>[12-14]</sup>, et al. presented some modifications with indicial response formulation at low Mach number. In this paper, the modified L-B state space model with state space formation at low Mach number is presented. The modified L-B model is also combined with airfoil structural dynamic equations of motion to present the airfoil aeroelastic system.

The purpose of this paper is to investigate airfoil aeroelastic phenomenon of stall flutter at low Mach number. In order to avoid the numerical integration

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issues of instability and inaccuracy, the implicit Newmark average velocity state space numerical integration method<sup>[15]</sup> is applied to solving the airfoil aeroelastic state space model. Based on bifurcation diagrams, phase plane plots and time domain response figures, the stall flutter behavior of airfoil aeroelastic system is investigated.

## 2. L-B State Space Model

### 2.1. General description of L-B model

The L-B model is represented by a set of ordinary differential equations (ODEs) of the form<sup>[10,16-17]</sup>

$$\dot{X} = f_u(X, \alpha, q), \quad \dot{X} = \frac{dX}{dt} \quad (1)$$

where  $f_u$  is the right hand terms of the L-B model's ODEs,  $X=[x_1 \ x_2 \ \dots \ x_n]$  the state space vector,  $n$  the number of aerodynamic model state variables,  $\alpha$  the angle of attack and  $q$  the non-dimensional pitch rate.

The airload coefficients of the airfoil are given by

$$C_i = g_i(X, \alpha, q) \quad i = N, m, C \quad (2)$$

where  $C_N$ ,  $C_m$  and  $C_C$  represent coefficients of normal force, pitching moment and chord force respectively.

Fig. 1 shows the definition of aerodynamic force in unsteady flow. The inflow velocity is denoted by  $V$ , and  $\alpha_E$  represents the effective angle of attack. Coefficients of lift force  $C_L$  and drag force  $C_D$  are given by

$$C_L = C_N \cos \alpha_E + C_C \sin \alpha_E \quad (3)$$

$$C_D = C_N \sin \alpha - C_C \cos \alpha \quad (4)$$

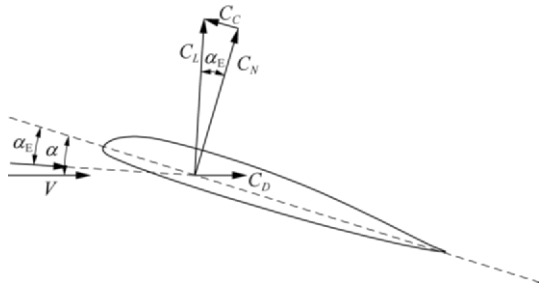


Fig. 1 Definition of aerodynamic force in unsteady flow.

Typically, the airload coefficients  $C_i$  are calculated as

$$C_N = C_N^I + C_N^f + C_N^v \quad (5)$$

$$C_m = C_m^I + C_m^f + C_m^v + C_{m0} \quad (6)$$

$$C_C = C_C^f \quad (7)$$

where the superscripts I, f and v refer to attached flow terms, trailing edge separation and dynamic stall (vortex induced) terms respectively, and  $C_{m0}$  is used for the break of the symmetry of airfoil.

### 2.2. Unsteady attached flow

Effective angle of attack  $\alpha_E$ , and the attached flow

parts of  $C_N$ ,  $C_m$  and  $C_C$  are defined from the state vector  $\tilde{x}=[x_1 \ x_2 \ \dots \ x_8]$  which is the solution of the following equation:

$$\dot{\tilde{x}} = A\tilde{x} + Bu \quad (8)$$

where  $u=[\alpha \ q]^T$ ,  $A$  and  $B$  are constant matrices from Ref. [16].

### 2.3. Separation flow

Under unsteady conditions, the leading edge pressure lag can be simply modeled by applying a first order lag to  $C_N^p$  ( $C_N^p$  is the normal force under potential flow conditions) to produce a substitute value  $C_N'$ . A new state space variable  $x_9$ , which represents  $C_N'$ , can be written in the form of a state space equation:

$$\dot{x}_9 = \left( -\frac{x_9}{T_p} + \frac{C_N^C + C_N^I}{T_p} \right) \frac{V}{b} \quad (9)$$

$$C_N^p = C_N^C + C_N^I \quad (10)$$

where  $T_p=1.7$  is the Mach number dependent constant<sup>[16]</sup>,  $b$  the semi-chord length,  $C_N^C$  the circulatory normal force.

Based on Kirchhoff theory, the static trailing edge separation point of  $f$  (as a fraction of the airfoil chord) is defined as

$$f(\alpha, \alpha_1) = \begin{cases} 1 - 0.3 \exp\left(\frac{|\alpha| - \alpha_1}{S_1}\right) & |\alpha| \leq \alpha_1 \\ 0.04 + 0.66 \exp\left(\frac{\alpha_1 - |\alpha|}{S_2}\right) & |\alpha| > \alpha_1 \end{cases} \quad (11)$$

where  $\alpha_1=15.25$ ,  $S_1=3.0$ , and  $S_2=2.3$  are Mach number dependent parameters<sup>[16]</sup>.

Under unsteady conditions, the trailing edge separation effects of  $C_N$ ,  $C_m$  and  $C_C$  are defined as the linear function of unsteady trailing edge separation point  $x_{10}$  which is given by

$$\dot{x}_{10} = \frac{f\left(\frac{x_9}{C_{N\alpha}}, \alpha_{1n}\right) - x_{10}}{\sigma_1 T_f} \cdot \frac{V}{b} \quad (12)$$

where  $C_{N\alpha}=0.108$  and  $T_f=3.0$  are Mach number dependent parameters<sup>[16]</sup>,  $\sigma_1$  the modification coefficient of time constant  $T_f$ , and  $\alpha_{1n}$  the function of  $x_{10}$ .

### 2.4. Dynamic stall

Eq. (13) shows the critical criterion for the onset of dynamic stall. A new state space variable  $x_{11}$ , which is used to track the position of the convected vortex, is solved by Eq. (14).

$$|C_N'| > C_{N1} \quad (13)$$

$$\dot{x}_{11} = \frac{V}{b} \quad (14)$$

where  $C_{N1}$  is Mach number dependent parameter.

A new state space variable  $x_{12}$ , which is introduced to represent vortex induced normal force  $C_N^v$ , is given by

$$\dot{x}_{12} = \begin{cases} \dot{c}_v - \frac{x_{12}}{\sigma_2 T_v} \cdot \frac{V}{b} & \alpha \dot{c}_v \geq 0 \text{ and } 0 < x_{11} < 2T_{v1} \\ -\frac{x_{12}}{\sigma_2 T_v} \cdot \frac{V}{b} & \text{Otherwise} \end{cases} \quad (15)$$

where  $T_v=6.0$  and  $T_{v1}=7.0$  are Mach number dependent parameters<sup>[16]</sup>,  $c_v$  represents the strength of vortex induced normal force, and  $\sigma_2$  is the modification coefficient of time constant  $T_v$ .

### 2.5. Flow reattachment

By introducing a new state space variable  $x_{13}$  which is the solution of Eq. (16) representing the trailing edge separation point during the flow reattachment phase, the pitching moment coefficient can be adjusted by  $x_{13}$  during this phase.

$$\dot{x}_{13} = \frac{f(\alpha, \alpha_{ln}) - x_{13}}{0.5T_f} \quad (16)$$

## 3. Modification at Low Mach Number

Though the L-B model is guided by the flow physics, it still requires some significant empirical constants which come from airfoil aerodynamic experiments. Beddoes supplied the empirical constants for L-B model with the Mach number ranging from 0.3 to 0.8, however, when compared with the airfoil airload experimental data of Glasgow University at low Mach number (<0.3), two main issues for predicting normal force existed in original L-B model: stall onset criterion is predicted too early and the overshoot of normal force coefficient after dynamic stall is not accurately calculated<sup>[11-14]</sup>. Based on state space formulation, the modified L-B model for the low Mach number is presented in the following sections.

### 3.1. Dynamic stall criterion modification

Under the conditions of low Mach number, an additional time lag is required within which the disturbed flow develops into vortex strong enough to cause the dynamic stall initiation. In order to explain this effect, the additional lagged value  $C_N''$ , which is the substitute value of  $C_N'$  under low Mach number conditions, is introduced in a similar manner to  $C_N'$ <sup>[10-14]</sup>. A new state space variable  $x_{14}$ , which represents  $C_N''$ , is the solution of the ODE of the system

$$\dot{x}_{14} = \frac{C_N' - x_{14}}{T_b} \cdot \frac{V}{b} \quad (17)$$

when  $C_N'' > C_{N1}$ , the dynamic stall initiates at low Mach number, where  $T_b=2.2$  is the time lag constant and  $C_{N1}=1.75$ <sup>[14]</sup>.

### 3.2. Modification of normal force and pitching moment coefficients

The vortex forms and detaches near the airfoil leading edge, and the flow at the separated shear layer still attaches to the upper surface at low Mach number. Hence, its effect causes the additional overshoot in normal force at low Mach number. Based on the suggestion of Beddoes<sup>[11]</sup>, the increased amplitude in normal force is proportional to the difference between the delayed separation point and its corresponding one at the steady state. Therefore, the overshoot for normal force coefficient  $\Delta C_N^v$  is given as follows:

$$\Delta C_N^v = B_1(f'' - f)V_x \quad (18)$$

where  $B_1=1.0$  is a coefficient related to airfoils;  $f''=x_{10}$  (as a function of airfoil chord) is the unsteady lagged trailing edge separation point;  $V_x$ , which represents the shape function of normal force due to vortex, is given by

$$V_x = \begin{cases} \sin^{3/2}\left(\frac{\pi\tau}{2T_v}\right) & 0 < \tau \leq T_v \\ \cos^2\left(\frac{\pi(\tau - T_v)}{T_{v1}}\right) & \tau > T_v \end{cases} \quad (19)$$

The additional pitching moment coefficient  $\Delta C_m^v$  is also the effect of vortex convection over the airfoil. It is given by

$$\Delta C_m^v = B_2\left(1 - \cos\frac{\pi\tau_v}{T_v}\right)\Delta C_N^v \quad (20)$$

where  $B_2=0.2$  is the coefficient dependent on airfoils.

## 4. Airfoil Aeroelastic Model

Fig. 2 shows the sketch of airfoil aeroelastic system. The airfoil vertical displacement is denoted by  $h$ ,  $c$  is the airfoil chord length, and  $\theta$  represents the airfoil pitch angle.  $a_h$  is the non-dimensional distance in semi-chord length from elastic axis to mid-chord, and  $x_\theta$  is the non-dimensional distance in semi-chord length of mass centre away from elastic axis.  $K_h$  and  $K_\theta$  are spring stiffness in bending and torsion. Based on the principle that the sum of inertia, elastic forces and moments must balance the externally applied force and moments, the airfoil aeroelastic model is presented as follows<sup>[18]</sup>.

$$m\ddot{h} + S\ddot{\theta} + hK_h = Q_h \quad (21)$$

$$S\ddot{h} + I_\theta\ddot{\theta} + \theta K_\theta = Q_\theta \quad (22)$$

where  $m$  is the mass per unit length,  $S$  the static mass moment,  $I_\theta$  is the polar moment of inertia about 1/4 chord,  $Q_h$  and  $Q_\theta$  are external aerodynamic force and moment respectively.

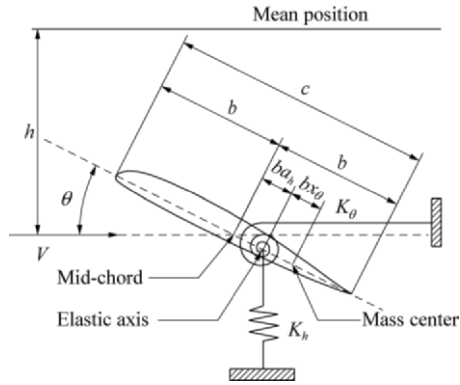


Fig. 2 Sketch of airfoil aeroelastic system.

The second order differential of Eq. (21) and Eq. (22) can be transferred into first order form by defining the state variables

$$[x_{15} \ x_{16} \ x_{17} \ x_{18}]^T = [h \ \theta \ \dot{h} \ \dot{\theta}]^T \quad (23)$$

The state space equations representing airfoil structural dynamic model are given by

$$\dot{\hat{x}} = A_s \hat{x} + B_s u_s \quad (24)$$

where  $A_s$  and  $B_s$  are coefficient matrices of airfoil structural dynamic model,  $\hat{x} = [x_{15} \ x_{16} \ x_{17} \ x_{18}]^T$

$$\text{and } u_s = \begin{bmatrix} 0 & 0 & -\frac{1}{2}\rho V^2 c C_L & \frac{1}{2}\rho V^2 c^2 C_m \end{bmatrix}^T.$$

The angle of attack  $\alpha$  and the non-dimensional pitch rate  $q$  can be written in terms of the structural state space variables:

$$\begin{bmatrix} \alpha \\ q \end{bmatrix} = \begin{bmatrix} \theta + \frac{h}{V} \\ \frac{\dot{\theta} c}{V} \end{bmatrix} = \begin{bmatrix} x_{16} + \frac{x_{17}}{V} \\ \frac{x_{18} c}{V} \end{bmatrix} \quad (25)$$

The structural state space Eq. (24) and modified L-B state space equations presented previously are coupled to complete the airfoil aeroelastic system with 18 state space variables.

## 5. Results and Analysis

### 5.1. Validation of modified L-B model at low Mach number

Based on the modified L-B model, the airload of NACA0012 is investigated. Figs. 3-4 show the normal force and pitching moment coefficients prediction

against angle of attack respectively when Mach number  $Ma=0.12$ , reduced frequency  $k=0.124$ ,  $c=0.55$ , and  $\alpha=15^\circ+10^\circ\sin(\omega t)$ , where  $\omega$  represents the circular frequency.

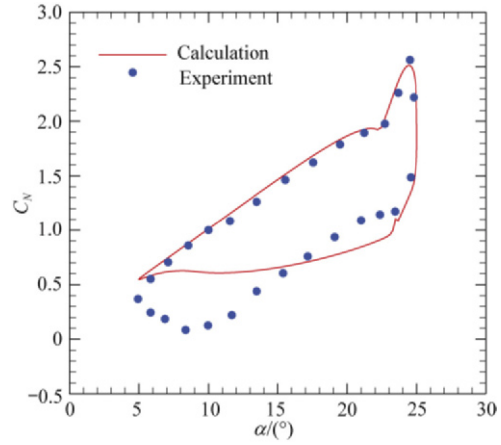


Fig. 3 NACA0012 normal force coefficient against angle of attack.

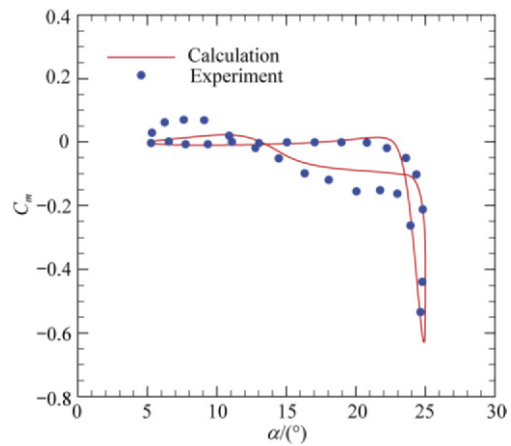


Fig. 4 NACA0012 pitching moment coefficient against angle of attack.

As shown in Fig. 3, the normal force coefficient predicted with modified L-B model at low Mach number is in good correlation with the experimental data<sup>[14]</sup>, the lagged dynamic stall onset pressure value and overshoot of normal force coefficient after dynamic stall initiation are accurately calculated, but the amplitude for normal force coefficient is a little higher than experimental data at the low angle of attack of flow reattachment phase, which makes the airfoil flutter analysis more conservative. Fig. 4 shows the pitching moment coefficient compared with the experimental result<sup>[14]</sup>, and the agreement between two curves is also fine.

In order to further validate the modified L-B model at low Mach number, the airload for OA207 airfoil is also investigated. As depicted in Figs. 5-6, with the oscillation state at  $\alpha=13^\circ+10^\circ\sin(\omega t)$ ,  $k=0.105$ ,  $V=34.24$  m/s and Reynolds numbers  $Re=2.0\times 10^6$ , the normal force and pitching moment coefficients of

modified L-B model are also well consistent with the experimental data<sup>[19]</sup>.

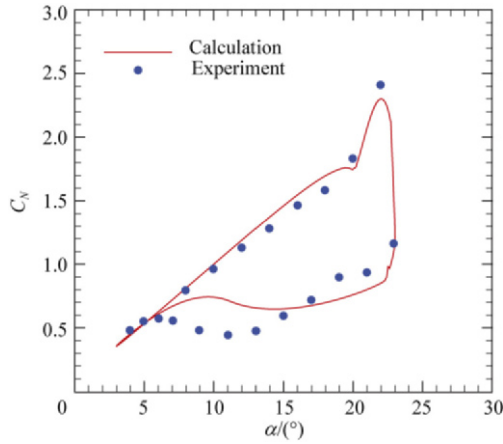


Fig. 5 OA207 normal force coefficient against angle of attack.

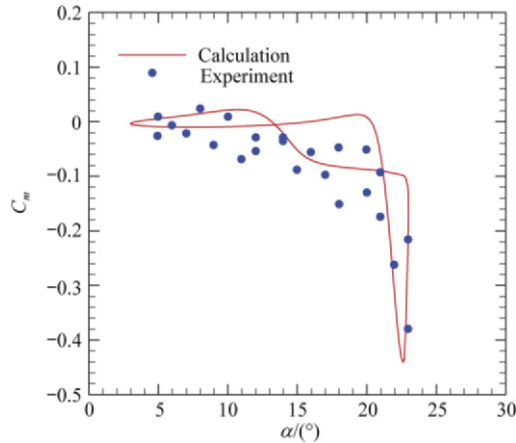


Fig. 6 OA207 pitching moment coefficient against angle of attack.

### 5.2. Airfoil stall flutter analysis at low Mach number

Ref. [3] presents a set of experiments conducted on NACA0012 airfoil undergoing the stall flutter oscillations at low Mach number, and they are chosen as the numerical example to verify and investigate the airfoil aeroelastic system mentioned previously. The main parameters of airfoil aeroelastic system are:  $c=0.3$  m,  $m=16.67$  kg/m,  $K_h=30.5$  N/mm,  $K_\theta=13.1$  N·m/rad and  $I_\theta=0.31$  kg/m<sup>2</sup>. The airfoil applied in this paper is symmetric, and the static pitching moment coefficient  $C_{m0}$  at zero angle of attack should be zero, but the static moment test curve of Ref. [3] shows that  $C_{m0}$  is non-zero at zero angle of attack, so in order to investigate the generic flutter behavior of asymmetric airfoil aeroelastic system<sup>[17,20]</sup>, the zero angle of attack static moment  $C_{m0}=0.01$  is applied for breaking the symmetry of the airfoil aeroelastic system<sup>[3]</sup>, and  $C_{m0}=0$ , 0.005 are also chosen for investigating the sensitivity of static moment coefficient to the airfoil aeroelastic system.

Figs. 7-9 present bifurcation diagrams of airfoil aeroelastic system against inflow velocity with static pitching moment coefficient of  $C_{m0}=0$ , 0.005, 0.010 respectively. Table 1 lists the predicted bifurcation behavior of airfoil aeroelastic system shown in Figs. 7-9 and the experimental flutter bifurcation behavior from Ref. [3]. Figs. 10(a)-10(c) show limit cycle oscillation (LCO) trajectories obtained from symmetric case

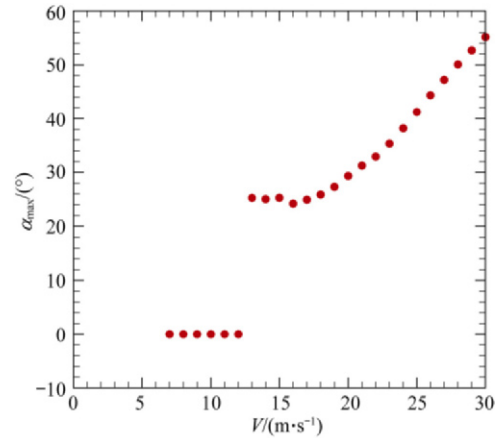


Fig. 7 Symmetric airfoil bifurcation diagram ( $C_{m0}=0$ ).

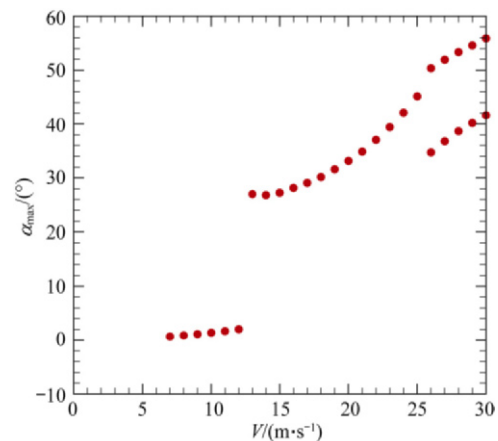


Fig. 8 Asymmetric airfoil bifurcation diagram ( $C_{m0}=0.005$ ).

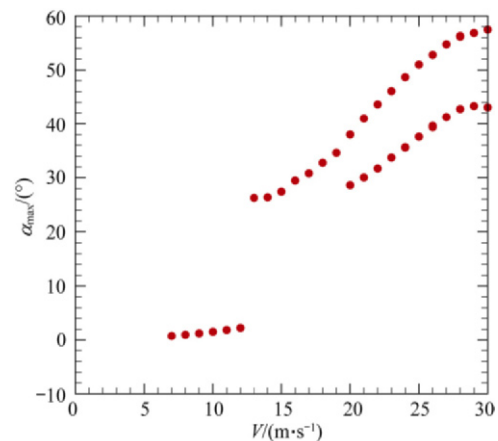
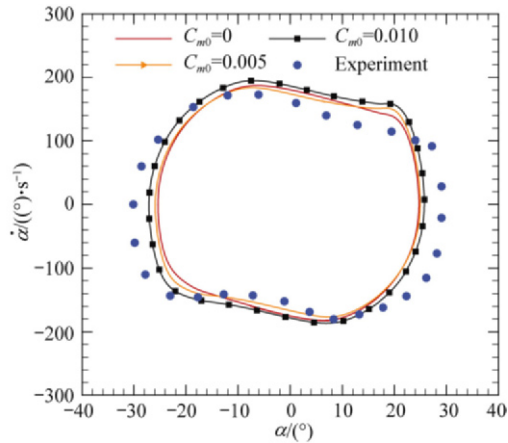
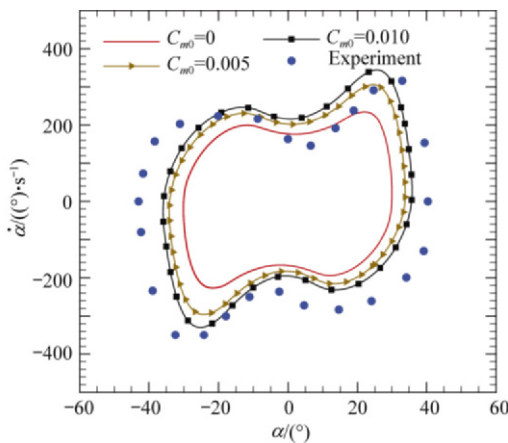
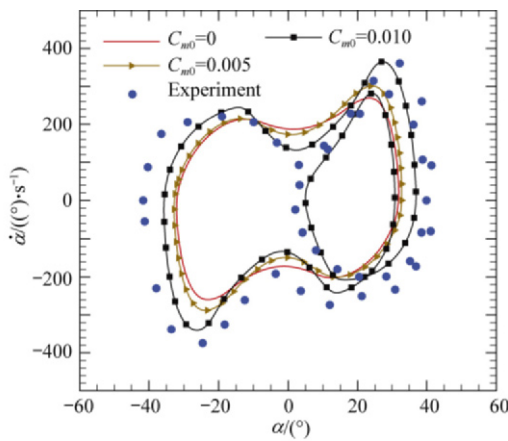


Fig. 9 Asymmetric airfoil bifurcation diagram ( $C_{m0}=0.010$ ).

**Table 1** Bifurcation behavior of airfoil flutter system

| Bifurcation motion case | $V/(m \cdot s^{-1})$ |                |                |                   |
|-------------------------|----------------------|----------------|----------------|-------------------|
|                         | $C_{m0}=0$           | $C_{m0}=0.005$ | $C_{m0}=0.010$ | Ref. [3]          |
| Decaying oscillation    | $<12$                | $<12$          | $<12$          | $<12$             |
| Hopf bifurcation        | $=12$                | $=12$          | $=12$          | $>12$ and $<13$   |
| Symmetric LCO           | $\geq 12$            | $\geq 12$      | $\geq 12$      | $>12$ and $<21.4$ |
| Asymmetric LCO          |                      | $>26$          | $>20$          | $>18.4$           |

(a)  $V=13$  m/s(b)  $V=19$  m/s(c)  $V=20$  m/sFig. 10 Phase plane between  $\alpha$  and  $\dot{\alpha}$ .

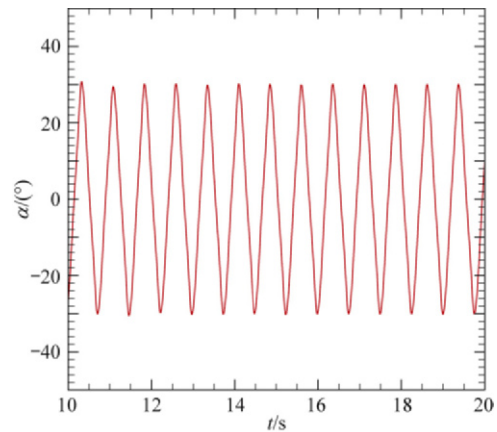
( $C_{m0}=0$ ), asymmetric cases ( $C_{m0}=0.005, 0.010$ ), and Ref. [3] with  $V=13, 19, 20$  m/s respectively. Based on Figs. 7-10 and Table 1, explanations for the bifurcation behavior of the airfoil aeroelastic system are as follows:

(1) When inflow velocity is less than 12 m/s, the oscillation of airfoil aeroelastic system decays to the equilibrium position. It is zero angle of attack for symmetric case ( $C_{m0}=0$ ), whereas for asymmetric cases ( $C_{m0}=0.005$  and  $0.010$ ), they are slightly above zero.

(2) When inflow velocity is beyond 12 m/s, the subcritical hopf bifurcation emerges for both symmetric and asymmetric cases, and the symmetric LCO happens suddenly. Obviously, as LCO trajectories shown in Fig. 10 at  $V=13$  m/s, the symmetric LCO trajectory of asymmetric case ( $C_{m0}=0.010$ ) is better correlated with experimental data, because  $C_{m0}=0.010$  is much more accurate to reflect the generic asymmetry of airfoil as previously described.

(3) As listed in Table 1 and shown in Figs. 7-9, the asymmetrical LCO emerges in two asymmetrical cases, but for symmetrical case, the asymmetrical LCO does not emerge. Also, seen from the comparison between Figs. 8-9, the asymmetric case with larger static moment coefficient ( $C_{m0}=0.010$ ) presents asymmetric LCO trajectory with lower inflow velocity ( $V=20$  m/s) which is closer to the experimental result listed in Table 1. Ref. [3] states that the asymmetric LCO happens after the airfoil static divergence velocity ( $V=17.7$  m/s), but the bifurcation diagrams of Figs. 7-9 conclude that the asymmetry of airfoil is another necessary condition for the emergence of asymmetric LCO.

Fig. 10(b) shows the LCO trajectories at  $V=19$  m/s, the airfoil aeroelastic system only presents symmetric LCO for both symmetric and asymmetric cases. Fig. 10(c) shows LCO trajectories at  $V=20$  m/s, symmetric and asymmetric LCOs coexist for the asymmetric case with  $C_{m0}=0.010$ , whereas for other two cases, the asymmetric LCO doesn't appear, and the similar conclusion can also be drawn from the comparisons among Fig. 11 which depicts the angle of attack re-

(a)  $C_{m0}=0$

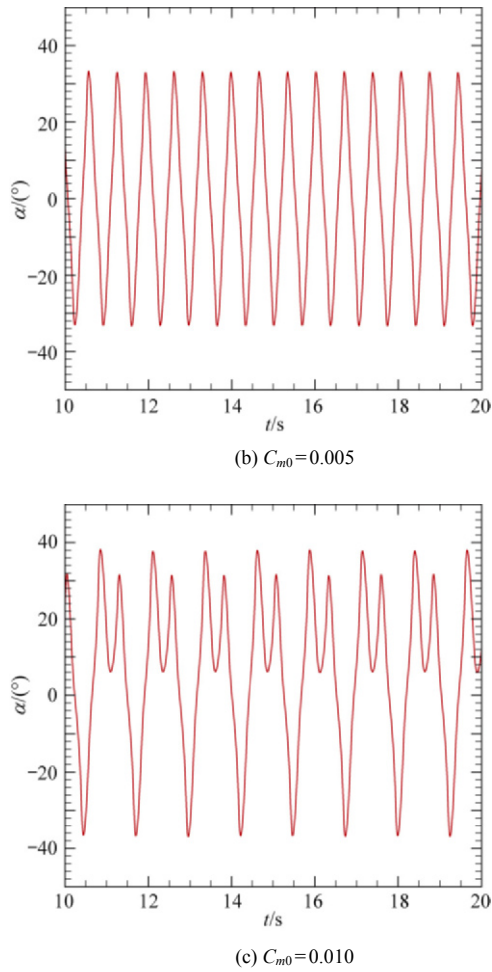
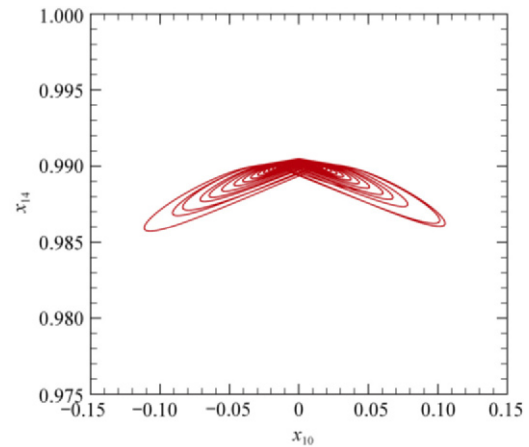


Fig. 11 Angle of attack response for asymmetric airfoil at  $V=20$  m/s.

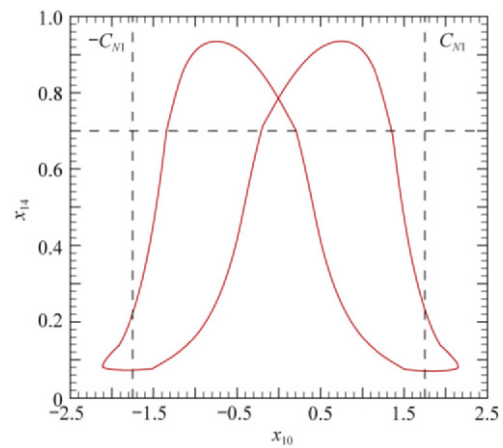
sponses for three cases ( $C_{m0}=0, 0.005, 0.010$ ) at  $V=20$  m/s respectively. As shown in Fig. 10, the LCO trajectory of asymmetric case with  $C_{m0}=0.010$  presents better correlation with experimental results of Ref. [3], which indicates that the static moment coefficient  $C_{m0}=0.010$  applied to investigating generic bifurcation behavior of airfoil aeroelastic system is suitable and accurate.

As described previously, the state space variables  $x_{10}$  and  $x_{14}$  represent the airfoil trailing separation point and the leading edge pressure coefficient respectively, which are applied to indicating dynamic stall initiation of L-B model. As shown in Fig. 12, the phase plane trajectories between  $x_{10}$  and  $x_{14}$  are used to investigate the stall flutter behavior for airfoil aeroelastic system with  $C_{m0}=0.010$  at  $V=12, 13, 20$  m/s respectively. Based on the airfoil aeroelastic system motion states presented in Table 1, when  $V=12$  m/s, the airfoil aeroelastic system tends to decay to equilibrium point. As clearly shown in Fig. 12(a), the phase plane trajectory is a retracting spiral curve, and the flow separation and the dynamic stall cannot be initiated. Fig. 12(b) shows the phase plane trajectory at  $V=13$  m/s. As mentioned above, the symmetric LCO presented as the result of subcritical hopf bifurcation emergence at this

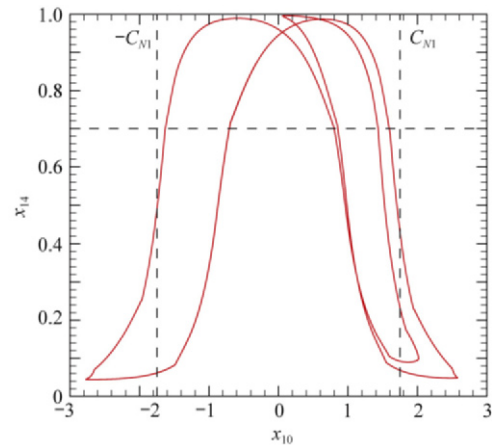
point. The phase plane trajectory crosses the vertical lines of  $x_{10}=\pm C_{N1}$ , which indicates that the flow separates on airfoil and the dynamic stall initiates. Hence, it can be concluded that the subcritical hopf bifurcation of airfoil aeroelastic system is caused by dynamic stall, which is defined as stall flutter<sup>[1-3]</sup>. At  $V=20$  m/s, two LCO trajectories (including an asymmetric LCO trajectory) coexist in the phase plane plot of Fig. 12(c), and stall flutter also happens at this point. But there is



(a)  $V=12$  m/s



(b)  $V=13$  m/s



(c)  $V=20$  m/s

Fig. 12 Phase plane between  $x_{10}$  and  $x_{14}$ .

no physical change from the viewpoint of dynamic stall, which reinforces that the asymmetric LCO is caused by the airfoil static divergence, and the same conclusion can be drawn from Ref. [3].

## 6. Conclusions

(1) Compared with experimental airfoil airload data under low Mach number conditions, it is shown that the modified L-B model with state space formulation is effective and accurate, and the implicit Newmark state space numerical integration method can effectively solve the issue of the numerical integration instability for the modified L-B state space model.

(2) The stall flutter behavior of airfoil aeroelastic system is sensitive to small changes in the value of static moment coefficient, and good correlation of results between the present model with suitable static moment coefficient and the experimental model validates the airfoil aeroelastic system in this paper.

(3) The physical mechanisms of LCOs are also investigated. The symmetric LCO emerges after subcritical hopf bifurcation caused by dynamic stall, and the asymmetric LCO, which is caused by the interaction of dynamic stall and static divergence, only happens in the asymmetric airfoil aeroelastic system.

## References

- [1] Lee B H K, Price S J, Wong Y S. Nonlinear aeroelastic analysis of airfoils: bifurcation and chaos. *Progress in Aerospace Sciences* 1998; 35(3): 205-334.
- [2] Sarkar S, Bijl H. Nonlinear aeroelastic behavior of an oscillating airfoil during stall-induced vibration. *Journal of Fluids and Structures* 2008; 24(6): 757-777.
- [3] Dimitriadis G, Li J. Bifurcation behavior of airfoil undergoing stall flutter oscillations in low-speed wind tunnel. *AIAA Journal* 2009; 41(11): 2577-2596.
- [4] Zhang Q C, Wang H L, Zhu Z W, et al. *Theory and application of bifurcation and chaos*. Tianjin: Tianjin University Press, 2005; 43-46. [in Chinese]
- [5] Bisplinghoff R L, Holt A, Robert L H. *Aeroelasticity*. New York: Dover Publications, 1996.
- [6] Leishman J G, Crouse G L. A state-space model of unsteady aerodynamics in a compressible flow for flutter analyses. *AIAA-1989-22*, 1989.
- [7] Price S J, Keleris J P. Non-linear dynamics of an airfoil forced to oscillate in dynamics stall. *Journal of Sound and Vibration* 1996; 194(2): 265-283.
- [8] Galvanetto U, Peiro J, Chantharasanawong C. Remarks on the nonlineardynamics of a typical aerofoil section in dynamic stall. *Aeronautical Journal* 2007; 111(1125): 731-739.
- [9] Seydel R. *Practical bifurcation and stability analysis*. 2nd ed. New York: Springer-Verlag, 1994.
- [10] Leishman J G, Crouse G L. State-space model for un-

steady airfoil behavior and dynamic stall. *Proceedings of the 30th AIAA/ASME/ASCE/AHS/ASC Structures*. 1989; 1319-1330.

- [11] Beddoes T S. A third generation model for unsteady aerodynamics and dynamic stall. Westland Limited RP-908, 1993.
- [12] Galbraith S W, McD R A, Coton F N. A new stall-onset criterion for low speed dynamic-stall. *Journal of Solar Energy Engineering* 2006; 128(4): 461-471.
- [13] Galbraith S W, McD R A, Coton F N. Improved dynamic stall onset criterion at low Mach numbers. *Journal of Aircraft* 2007; 44(3): 1049-1052.
- [14] Galbraith S W, McD R A, Coton F N. A modified dynamic stall model for low Mach numbers. *Journal of Solar Energy Engineering* 2008; 130(3): 31013-31022.
- [15] Kim B O, Lee A S. A transient response analysis in the state-space applying the average velocity concept. *Journal of Sound and Vibration* 2005; 281(3-5): 1023-1035.
- [16] Bauchau O A. *DYMORE user's and theory manual*. <<http://www.ae.gatech.edu/people/obauchau/>>. Atlanta, Georgia: Georgia Institute of Technology, Daniel Guggenheim School of Aerospace Engineering, 2007.
- [17] Chantharasanawong C. *Nonlinear aeroelastic behaviour of aerofoils under dynamic stall*. PhD thesis, University of London, 2007.
- [18] Fung Y C. *An introduction to the theory of aeroelasticity*. London: Chapman & Hall, 1993; 210-212.
- [19] Tan J F. *Analysis of rotor aerodynamic response under manoeuvring conditions*. Master thesis, Nanjing University of Aeronautics and Astronautics, 2009. [in Chinese]
- [20] Galvanetto U. Some remarks on the two-block symmetric Burridge-Knopoff. *Physics Letter. A* 2002; 293(5-6): 251-259.

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